Year 9 End of Year Biology Checklist 2026

Topic	Student Checklist	😊	☹️
4.1.1 Cell structure	Use the terms 'eukaryotic' and 'prokaryotic' to describe types of cells		
	Describe the features of bacterial (prokaryotic) cells		
	Demonstrate an understanding of the scale and size of cells and be able to make order of magnitude calculations, including standard form		
	Recall the structures found in animal and plant (and algal) cells		
	Use estimations and explain when they should be used to judge the relative size or area of sub-cellular structures		
	Required practical 1: use a light microscope to observe, draw and label plant and animal cells		
	Describe the functions of the structures in animal and plant (eukaryotic) cells		
	Describe what a specialised cell is, including examples for plants and animals		
	Describe what differentiation is, including differences between animals and plants		
	Define the terms magnification and resolution		
	Compare electron and light microscopes in terms of their magnification and resolution		
	Carry out calculations involving magnification using the formula: magnification = size of image/ size of real object (including standard form)		
4.1.2 Cell division	Describe how genetic information is stored in the nucleus of a cell (genes & chromosomes)		
	Describe the processes that happen during the cell cycle, including mitosis (recognise and describe where mitosis occurs)		
	Describe stem cells, including sources of stem cells in plants and animals and their roles		
	Describe the use of stem cells in the production of plant clones and therapeutic cloning		
	Discuss the potential risks, benefits and issues with using stem cells in medical research/treatments (including the examples of treating diabetes and paralysis)		
4.1.3 Transport in cells	Describe the process of diffusion, including examples		
	Explain how diffusion is affected by different factors		
	Define and explain "surface area to volume ratio", and how this relates to single-celled and multicellular organisms (including calculations)		
	Explain how the effectiveness of an exchange surface can be increased, including examples of adaptations for small intestines, lungs, gills roots & leaves		
	Describe the process of osmosis (including calculation of water uptake & percentage gain and loss of mass of plant tissue)		
	Required practical 3: Osmosis experiment: Investigate the effect of a range of concentrations of salt or sugar solutions on the mass of plant tissue		
	Describe the process of active transport, including examples - gut and roots		
	Explain the differences between diffusion, osmosis and active transport		

4.2.1 Principles of organisation & 4.2.2 Animal tissues, organs and organ	Describe the levels of organisation within living organisms (cell<tissue<organ<organ system)		
	Describe the digestive system and how it works as an organ system (from KS3)		
	Describe basic features of enzymes, including rate calculations (…. ÷ time)		
	Describe the lock and key theory as a model of enzyme action and explain how the shape a of the active sites makes the enzyme specific		
	Explain the effect of temperature and pH on enzymes		
	Describe the digestive enzymes, including their names, sites of production and actions		
	Describe how the products of digestion are used		
	Describe the features and functions of bile and state where it is produced and released from		
	Required practical 4: Food Tests: use qualitative reagents to test for a range of carbohydrates, lipids and proteins		
	Required practical 5: investigate the effect of pH on the rate of reaction of amylase enzyme		
	Describe the structure of the lungs including how lungs are adapted for gas exchange		